Olist Commerce360

An End-to-End E-Commerce Data Warehouse and Business Intelligence Platform

P1_02 - Business Requirements Document

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Submission date	25 October 2026

1. Purpose

This document records who needs information from the Olist Commerce360 warehouse, which business processes must be modelled, the business requirements and KPI definitions, and the data and non-functional requirements that the implementation (Phases 2 and 3) must satisfy. It is the reference for the business questions (P1_03) and the dimensional design (P1_04 to P1_07).

2. Stakeholders and information needs

Stakeholder	Information need	Delivered through
Executive management	Overall sales, order and customer trends; regional growth	Executive Overview dashboard page
Category managers	Category and product revenue, units, freight burden	Product analysis
Marketplace / seller relations	Seller contribution and concentration; seller locations	Seller analysis
Logistics / operations	Order status, fulfilment stage times, on-time delivery by region	Delivery performance analysis
Customer experience	Review scores, low-score share, link to delivery	Satisfaction analysis
Finance	Payment method mix, amounts paid, instalments	Payment analysis
Marketing / CRM	New vs repeat customers, customer geography	Customer analysis
BI / data team (project team)	Correct, reconciled, documented and reproducible data	Warehouse, ETL, data-quality reports

3. Business processes

A data warehouse is organised around measurable business events, not around the source tables. Four processes produce measurable events in the source data:

Process	Business event	Source	Fact table	Fact type
Sales (order line items)	A customer buys one or more products; each product line is sold by one seller.	order_items + orders	fact_order_items	Transaction
Order fulfilment lifecycle	An order moves through purchase, payment approval, carrier hand-off and delivery against an estimated date.	orders (+ per-order aggregates)	fact_orders	Accumulating snapshot
Payment transactions	The customer pays for the order using one or more payment records (card, boleto, voucher).	order_payments + orders	fact_payments	Transaction
Customer reviews	After delivery (or the estimated date) the customer is sent a survey and may give a 1-5 score and a comment.	order_reviews + orders	fact_reviews	Transaction

Each of these processes has a different number of rows per order. Per order, the verified maximum is 21 item rows, 29 payment rows and 3 review rows. For this reason each process gets its own fact table.

4. Business requirements

ID	Requirement	Stakeholder	Process	Priority
BR-01	Track sales revenue, units and orders over time (month, quarter, year) with growth rates.	Executive management	Sales	Must
BR-02	Analyse sales by product category and product.	Category managers	Sales	Must
BR-03	Analyse sales and seller base by geography (region, state, city) for both customers and sellers.	Executive management, Marketplace operations	Sales, Order fulfilment	Must
BR-04	Measure seller contribution and concentration of revenue.	Marketplace / seller relations	Sales	Must
BR-05	Report payment method mix, amounts paid and instalment behaviour.	Finance	Payments	Must
BR-06	Monitor order status distribution and cancellations.	Operations	Order fulfilment	Must
BR-07	Measure delivery performance: lead times per fulfilment stage and delivery against the promised date.	Logistics / operations	Order fulfilment	Must
BR-08	Monitor customer satisfaction through review scores and comments.	Customer experience	Reviews	Must
BR-09	Relate delivery performance to customer satisfaction at order level.	Customer experience, Logistics	Order fulfilment, Reviews	Must
BR-10	Identify new versus repeat customers and their contribution to sales.	Marketing / CRM	Order fulfilment, Sales	Must
BR-11	Compare freight cost relative to item price across categories and regions.	Logistics, Category managers	Sales	Should
BR-12	All figures must reconcile to the source files (row counts and value totals) and use one agreed definition per KPI.	BI / data team	All	Must

5. KPI definitions

Each KPI has one agreed definition. Item sales revenue, freight, gross merchandise value and amount paid are different measures and are never used interchangeably. Unless stated otherwise, revenue KPIs exclude orders in the *Cancelled* status group (canceled and unavailable). In the source, these orders hold 549 item lines worth BRL 97,242.96.

KPI	Definition	Fact	Calculation	Additivity
Item sales revenue	Value of goods sold, excluding freight.	fact_order_items	SUM(item_price) WHERE status_group <> 'Cancelled'	Additive
Freight revenue	Freight charged to customers.	fact_order_items	SUM(freight_value) WHERE status_group <> 'Cancelled'	Additive
Gross merchandise value (GMV)	Item price plus freight.	fact_order_items	SUM(item_gross_value) WHERE status_group <> 'Cancelled'	Additive
Units sold	Number of items sold.	fact_order_items	SUM(item_quantity)	Additive
Order count	Number of distinct orders.	fact_orders	COUNT(*) on fact_orders	Additive
Average order value (AOV)	Item value per non-cancelled order.	fact_orders	SUM(order_item_value) / COUNT(*) WHERE status_group <> 'Cancelled' AND item_count > 0	Non-additive (ratio)
Amount paid	Money received from customers across all payment methods.	fact_payments	SUM(payment_value)	Additive
Payment count	Number of payment records.	fact_payments	COUNT(*) on fact_payments	Additive
Average instalments	Mean instalments per payment record.	fact_payments	AVG(payment_installments)	Non-additive
Cancellation rate	Share of orders cancelled or unavailable.	fact_orders	COUNT(status_group = 'Cancelled') / COUNT(*)	Non-additive (ratio)
On-time delivery rate	Share of delivered orders that arrived on or before the estimated date.	fact_orders	1 - SUM(is_late) / SUM(is_delivered)	Non-additive (ratio)
Average delivery days	Mean days from purchase to delivery.	fact_orders	AVG(days_purchase_to_delivery)	Non-additive
Average delivery delay	Mean days delivered after (+) or before (-) the estimate.	fact_orders	AVG(delivery_delay_days)	Non-additive
Repeat customer rate	Share of customers with more than one order.	fact_orders	COUNT(DISTINCT customer_key WHERE is_repeat_purchase = 1) / COUNT(DISTINCT customer_key)	Non-additive (ratio)
Average review score	Mean satisfaction score.	fact_reviews	AVG(review_score)	Non-additive

KPI	Definition	Fact	Calculation	Additivity
Low-score share	Share of reviews scoring 1 or 2.	fact_reviews	COUNT(review_score <= 2) / COUNT(*)	Non-additive (ratio)
Freight-to-price ratio	Freight as a share of item price.	fact_order_items	SUM(freight_value) / SUM(item_price)	Non-additive (ratio)

6. Data requirements

The source files and their verified properties are listed in the Project Proposal (P1_01, section 5) and in the Source_Profile sheet of the data dictionary (P1_07). The data-quality issues below were found during profiling. The warehouse must handle each one explicitly:

ID	Finding (verified)	Required handling
DR-01	customer_id is unique per order; customer_unique_id identifies the person (99,441 vs 96,096).	Customer dimension keyed on customer_unique_id.
DR-02	Zip prefixes have leading zeros ('01037') that are lost when read as integers.	Load as text, pad to 5 digits.
DR-03	610 products without category; 2 categories missing from the translation file.	'uncategorized' value; manual translation of 2 names.
DR-04	Approval (160), carrier (1,783) and delivery (2,965) timestamps are NULL, mostly for undelivered orders; 8 delivered orders have no delivery date.	Unknown date key -1; delivery measures NULL; excluded from delivery KPIs.
DR-05	1,359 orders have carrier date earlier than approval date; 23 have delivery earlier than carrier date.	Flag in data-quality report; exclude negative stage durations from averages.
DR-06	review_id is not unique (814 duplicates); 547 orders have more than one review.	Review fact keyed on review_id + order_id.
DR-07	Geolocation: 1,000,163 rows, 261,831 exact duplicates, coordinates outside Brazil.	Median coordinates per zip prefix inside Brazil's bounds.
DR-08	775 orders have no items (mostly unavailable/canceled); 1 order has no payment record.	Kept in fact_orders with zero item values / NULL payment.
DR-09	Amount paid equals item price + freight within BRL 0.01 for 99.69% of orders.	Amount paid is reported separately from sales revenue.
DR-10	Orders per month: 4 in Sep 2016, 1 in Dec 2016, 16 in Sep 2018, 4 in Oct 2018.	Trend analysis restricted to Jan 2017 - Aug 2018.

7. Non-functional requirements

ID	Requirement
NFR-01	Reproducibility: every load step is a committed SQL or Python script; the raw CSVs can be re-downloaded from Kaggle.
NFR-02	Reconciliation: fact row counts and SUM(price), SUM(freight_value), SUM(payment_value) equal the source totals (BRL 13,591,643.70; 2,251,909.54; 16,008,872.12).
NFR-03	Referential integrity: zero fact rows without a matching dimension row (unknown member -1 used instead of NULL).
NFR-04	Data volume: at least 100 dimension records and 10,000 fact records (expected: 96,097 customers; 112,650 item facts).
NFR-05	Performance: indexes on fact foreign keys and date keys; before/after EXPLAIN ANALYZE evidence (Phase 3).
NFR-06	Usability: Power BI imports separate fact and dimension tables with single-direction 1:* relationships, not one flattened table.
NFR-07	Security: no credentials in the repository; connection settings kept in a local, git-ignored file.
NFR-08	Documentation: table and column names identical across the diagram, data dictionary and documents.

8. Assumptions and constraints

- The Kaggle extract (version 2) is a static snapshot. The order status is the final status at extraction time; no status history exists.
- Monetary values are in Brazilian reais (BRL). No currency conversion is applied.
- Each order_items row is one unit. Repeated rows for the same product mean a quantity greater than one.
- Customer location is taken from the most recent order (SCD Type 1). Only 250 of 96,096 persons have more than one zip prefix.
- Analyses show associations only. For example, a relationship between delivery delay and review score is not presented as cause and effect.

9. Traceability

Each requirement is answered by at least one business question and fact table:

Requirement	Business questions	Fact tables
BR-01	BQ-01	fact_order_items, fact_orders
BR-02	BQ-02	fact_order_items
BR-03	BQ-03	fact_orders
BR-04	BQ-04	fact_order_items
BR-05	BQ-05, BQ-06	fact_payments
BR-06	BQ-07	fact_orders
BR-07	BQ-08, BQ-09	fact_orders
BR-08	BQ-10	fact_reviews
BR-09	BQ-11	fact_orders
BR-10	BQ-12	fact_orders
BR-11	BQ-13	fact_order_items
BR-12	All (reconciliation, P1_05 section 6)	All